

Shifan Xu

PH.D. CANDIDATE IN APPLIED PHYSICS

Yale University, New Haven, CT

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Education

Yale University

PH.D. CANDIDATE IN APPLIED PHYSICS

Advisors: Prof. Yongshan Ding and Prof. Steven M. Girvin

New Haven, CT, USA

Sep. 2021 – Present

University of Chicago

VISITING PH.D. STUDENT; HOST: PROF. LIANG JIANG

Chicago, IL, USA

Jan. 2024 – May 2024

Nanjing University

B.S. IN PHYSICS

Nanjing, China

Sep. 2017 – Jun. 2021

University of California, Berkeley

VISITING UNDERGRADUATE STUDENT

Berkeley, CA, USA

Jul. 2019 – Jan. 2020

Research Experience

Yale University

DOCTORAL RESEARCH; GRADUATE RESEARCH ASSISTANT

New Haven, CT

Sep. 2021–Present

- **Quantum memory systems.** Designed system-level and high-bandwidth QRAM architectures and developed supporting simulators.
- **Fault-tolerant quantum architecture.** Developed magic-state preparation methods and contributed to qLDPC error correction, routing, and logical operations.
- **Current directions.** Error-corrected quantum memory, full-stack qLDPC resource estimation, and AI-assisted architecture design.

Pacific Northwest National Laboratory

GRADUATE STUDENT INTERN; MENTOR: DR. ANG LI

Richland, WA

Jun.–Aug. 2025

- Researched oblivious QRAM for secure delegated quantum queries, leading to the co-first-author OQRAM preprint.

Publications

PEER-REVIEWED PUBLICATIONS AND ACCEPTED PAPERS

Shifan Xu, Kun Liu, Patrick Rall, Zhiyang He, and Yongshan Ding. “Distilling Magic States in the Bicycle Architecture.” In Proceedings of the 2026 ACM/IEEE 53rd Annual International Symposium on Computer Architecture (ISCA), pp. 2081–2095. 2026. DOI: 10.1109/ISCA66397.2026.00148 · arXiv:2602.20546.

Kun Liu, Takahiro Tsunoda, Sophia H. Xue, Evan McKinney, Zeyuan Zhou, **Shifan Xu**, Robert J. Schoelkopf, and Yongshan Ding. “Efficient Routing of Quantum LDPC Codes on Programmable 2D Toric Architectures.” Accepted to the IEEE International Conference on Quantum Computing and Engineering (QCE), 2026. arXiv:2604.18714.

Shifan Xu, Alvin Lu, and Yongshan Ding. “Fat-Tree QRAM: A High-Bandwidth Shared Quantum Random Access Memory for Parallel Queries.” Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), Volume 2. 2025. DOI: 10.1145/3676641.3716256 · arXiv:2502.06767.

Samuel Stein, **Shifan Xu**, Andrew W. Cross, Theodore J. Yoder, Ali Javadi-Abhari, Chenxu Liu, Kun Liu, Zeyuan Zhou, Charles Guinn, Yufei Ding, Yongshan Ding, and Ang Li. “HetEC: Architectures for Heterogeneous Quantum Error Correction Codes.” Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**), Volume 2. 2025. DOI: 10.1145/3676641.3716001 · arXiv:2411.03202.

Shifan Xu, Connor T. Hann, Ben Foxman, Steven M. Girvin, and Yongshan Ding. “Systems Architecture for Quantum Random Access Memory.” In Proceedings of the 56th Annual IEEE/ACM International Symposium on Microarchitecture (**MICRO**), pp. 526–538. 2023. DOI: 10.1145/3613424.3614270 · arXiv:2306.03242.

PREPRINTS

Shifan Xu^{*}, Yizhuo Tan^{*}, Yongshan Ding, and Jakub Szefer. “OQRAM: Oblivious Quantum Random Access Memory for Securing Delegated Quantum Queries.” arXiv preprint arXiv:2607.27171 (2026). ^{*}Co-first authors.

Dantong Li, **Shifan Xu**, and Yongshan Ding. “Asymptotically Optimal Depth Fermionic Permutation on 2D Grid Quantum Architecture without Ancillas.” arXiv preprint arXiv:2605.26041 (2026).

Kun Liu, **Shifan Xu**, Tomas Jochym-O’Connor, Zhiyang He, Shraddha Singh, and Yongshan Ding. “In-Situ Simultaneous Magic State Injection on Arbitrary CSS qLDPC Codes.” arXiv preprint arXiv:2604.05126 (2026).

Adrian Harkness, Shuwen Kan, Chenxu Liu, Meng Wang, John M. Martyn, **Shifan Xu**, Diana Chamaki, Ethan Decker, Ying Mao, Luis F. Zuluaga, Tamás Terlaky, Ang Li, and Samuel Stein. “FTCircuitBench: A Benchmark Suite for Fault-Tolerant Quantum Compilation and Architecture.” arXiv preprint arXiv:2601.03185 (2026).

D. K. Weiss, **Shifan Xu**, Shruti Puri, Yongshan Ding, and Steven M. Girvin. “Faulty towers: recovering a functioning quantum random access memory in the presence of defective routers.” arXiv preprint arXiv:2411.15612 (2024).

MANUSCRIPT IN PREPARATION

Shifan Xu, Connor T. Hann, Nathan Wiebe, Steven M. Girvin, and Yongshan Ding. “Error Corrected QRAM Queries with One-Fault-Tolerant CSWAP Gates.” 2026.

Research Software & Tutorials

PennyLane / Xanadu collaboration. Contributed QRAM implementations to PennyLane and coauthored the public tutorial below, comparing select-only, bucket-brigade, and hybrid QRAM constructions through executable examples and resource analysis. PennyLane 0.44 release announcement.

Shifan Xu, David Ren, Marcus Edwards, and Yongshan Ding. *Intro to quantum random access memory (QRAM)*. PennyLane Demos, August 30, 2026. Tutorial and code.

QRAM simulators. Developed simulation software for quantum random-access memory architectures.

Technical Skills

Programming languages: Python, C.

Research tools: QRAM simulators; quantum circuit simulation and resource estimation with PennyLane.

Awards & Fellowships

- 2020 **ZhengGang Fellowship**, Nanjing University
- 2019 **People Scholarship**, Nanjing University
- 2019 **Outstanding Prize in Taiwan College Physics Experiment Tournament (TCPT)**, National Tsing Hua University
- 2018 **Outstanding Prize in Top-notch Talent Plan**, Nanjing University

Teaching Experience

TEACHING ASSISTANTSHIPS

Yale University

New Haven, CT, USA

GRADUATE TEACHING ASSISTANT

Fall 2026

- CPSC 4310/5310: Computer Music: Algorithmic and Heuristic Composition
- Instructor: Scott Petersen

Yale University

New Haven, CT, USA

GRADUATE TEACHING ASSISTANT

2022 – 2023

- CPSC 4470/5470: Introduction to Quantum Computing
- Instructor: Prof. Yongshan Ding

GUEST LECTURES

Mar. 2026. *Introduction to Quantum Query Architecture*. CPSC 4490/5490: Quantum Information Systems, Yale University. Host: Prof. Yongshan Ding.

Feb. 2025. *Introduction to Quantum Query Architecture*. CSCI 4965, Rensselaer Polytechnic Institute. Host: Prof. Zhiding Liang.

Mentoring

Alvin Lu

New Haven, CT, USA

UNDERGRADUATE STUDENT, YALE UNIVERSITY

2024 – Present

- Mentored research on high-bandwidth shared quantum memory, leading to the coauthored paper *Fat-Tree QRAM* at ASPLOS 2025.

Presentations

INVITED TALKS

Apr. 2026. *Distilling Magic States in the Bicycle Architecture*. Invited talk hosted by the Xanadu team.

Spring 2024. *Architectural Improvements in Quantum Random Access Memory*. Invited talk at Prof. Liang Jiang's Group Meeting, University of Chicago.

Fall 2023. *Systems Architecture for Quantum Random Access Memory*. Invited talk at QAISG's Quantum Machine Learning Seminar, Centre for Quantum Technologies, National University of Singapore.

Professional Service

PROGRAM COMMITTEE

2024. Quantum Computing Applications and Systems Workshop, co-located with the 43rd IEEE/ACM International Conference on Computer-Aided Design (ICCAD). New Jersey, USA.

JOURNAL REFEREEING

Physical Review Research

Physical Review A (**PRA**)

PRX Quantum

Physical Review Letters (**PRL**)

Physical Review X (**PRX**)

IEEE Transactions on Parallel and Distributed Systems (**TPDS**)

ACM Transactions on Design Automation of Electronic Systems (**TODAES**)

ACM Transactions on Quantum Computing (**TQC**)

CONFERENCE REVIEWING

Quantum Information Processing (**QIP**)

Earlier Research Experience

Nanjing University - Physics Department

Nanjing, China

ADVISOR: DR. TIANJUE MIN

2021

- Graduation Thesis Project: "Analysis of Background Noise in $p\bar{p}$ Collision Experiments Based on the K-Nearest Neighbors (KNN) Algorithm"

UC San Diego - Department of NanoEngineering

(remote) San Diego, CA, USA

ADVISOR: DR. SHENG XU

2020

- Project: Circuit Design for Flexible Ultrasonic Array

Nanjing University - Engineering Department

Nanjing, China

ADVISOR: DR. XIN LI, DR. MINGHUI LU

2019-2021

- Project: Preparation and Design of Auxetic Metamaterials

Nanjing University - Physics Department

Nanjing, China

SUPERVISOR: PROF. SIHUI WANG

2018-2019

- Project: Metal Powder Spectra Analysis